LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 2: Events and Parallel Scripts

Module 1 | 45 minutes | Event-driven programming | Grades 6–8 · ages 11–14

Objective: Use event hats (flag, key, click, broadcast) to make programs respond to users, and run several scripts at the same time.

Success criteria

- I can use event hats to respond to users.
- I can run scripts in parallel.
- I can make sprites communicate with broadcasts.

Key vocabulary

- **event**: Something that happens, such as a click or key press.
- **hat block**: A block that starts a script when its event happens.
- **parallel**: Running at the same time.
- **broadcast**: A message sent to every sprite.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Event cards (5 min): learners hold cards like “when clap → stand up”; the teacher triggers events.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Drone controls: learners follow the steps.
14 min	**Perform**	Code Lab: drone controls, Code Lab: broadcast a signal, Which event starts each script?
9 min	**Probe**	activity
4 min	**Wrap-up**	Exit ticket: Which block lets one sprite tell another to start?

Unplugged warm-up

Event cards (5 min): learners hold cards like “when clap → stand up”; the teacher triggers events.

Teaching the idea

Event-driven programming: scripts are triggered by events (flag, key, click, broadcast). Multiple scripts run concurrently. Broadcasts decouple sender and receiver. Links to JavaScript event handlers.

Analogy: A fire station: firefighters wait until the alarm (event) rings, then act.

Live demo: Show two scripts on one sprite running at once (spin forever + move with keys).

Questions to ask

Ask	Listen for
What starts a script in Scratch?	A hat block event.
Can two scripts run at once?	Yes — in parallel.
Who receives a broadcast?	Every sprite with a matching receive hat.

Differentiation

Support
Provide the right-arrow script; learners copy and adapt.

Stretch
Add diagonal moves with combinations, or a “pause” broadcast.

Common misconceptions

Misconception	How to fix it
Scripts run one after another	Each hat starts its own script in parallel.
Broadcast goes to one sprite	It goes to all sprites.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which block lets one sprite tell another to start?

Homework
List five events that a smartphone responds to.

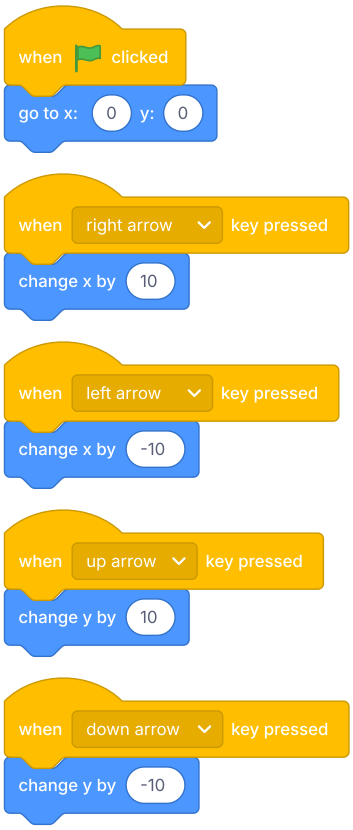
Answer key — Kitabash activities

1. Code Lab: drone controls (perform · code lab)

Goals checked: → moves right · ← moves left · ↑ moves up · ↓ moves down · Resets on the flag

One solution:

Drone



2. Code Lab: broadcast a signal (perform · code lab)

Goals checked: Click sends “party” · Robot reacts · Robot listens

One solution:

Drone



Robot



3. Which event starts each script? (perform · categorize)

Green flag: Reset the game at the start
Key press: Jump when space is pressed, Move left with the arrow
Sprite click: Open a menu when a button sprite is tapped
Broadcast: All enemies freeze when “pause” is sent

4. Explain your thinking (probe · free response)

Look for: event, when, happens. **Example:** A music app waits for events. When I tap play, the song starts. When a song ends, it gets an event and plays the next one automatically.

Answer key — worksheet

1. **Key words:** event = Something that happens, such as a click or key press. · hat block = A block that starts a script when its event happens. · parallel = Running at the same time. · broadcast = A message sent to every sprite.
2. **Plan your code:** Goals: → moves right; ← moves left; ↑ moves up; ↓ moves down; Resets on the flag.
3. **Plan your code:** Goals: Click sends “party”; Robot reacts; Robot listens.
4. **Which event starts each script?:** Green flag: Reset the game at the start · Key press: Jump when space is pressed, Move left with the arrow · Sprite click: Open a menu when a button sprite is tapped · Broadcast: All enemies freeze when “pause” is sent
5. **Explain your thinking:** A music app waits for events. When I tap play, the song starts. When a song ends, it gets an event and plays the next one automatically.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.